Quick Snapshots of Cambridge in Artist-Eye

Wenzhao Li*
Department of Computer Science
University of Cambridge
wl301@cam.ac.uk

Abstract

Neural style transfer is a computer vision technology that recomposes an image's content in another's style using a neural network. This technology makes it possible that a photograph might appear as if it were created by a famous artist. Nowadays, neural style transfer is extensively used in real-time photo and video editing software. However, this is not achieved overnight. Numerous scholars have explored how to improve the speed and size of the model. In this project, we want to have a taste of how these improvements happen in practice. We select two breakthrough methods in this field to implement - 1. perceptual losses feed-forward transformation network for neural style transfer speed-up and 2. model pruning for compression. We create and train our stylization model - Artist Eye - and explore how the ordinary view of Cambridge would be transformed into various artist styles. When compared with the baseline, our trained and pruned model achieves 118x faster in speed and 53x smaller in size, without obvious visual performance degradation. Besides qualitative comparison, we conduct quantitative analysis by using the state-of-the-art metric ArtFid. This project is a record of our discovery and hands-on practice process in this field. We also include our observations and findings in the Experiments and Evaluation Section.

Project link: click here

1 Introduction

Artists see the world as a place full of possibilities rather than as a set, unchanging paradigm for how to live. They are able to perceive much of what is invisible to the majority of people because they have less limitations on their thinking. Their unique way of interpreting objects makes most ordinary people feel miraculous and amazed. Most of artists live in the same environment and see the same scenery as us, but they seem to have magic - somehow inject their ideas into the artworks and create masterpieces by using various art style.

The neural style transfer technique [1] unravels the mysteries of the artist. With the aid of this technology, a photograph may look to have been produced by a well-known artist. The application of neural style transfer in real-time photo editing software has become widespread in recent years. To simulate the practical use of neural style transfer in daily photo editing, we collect a dataset which contains photos of everyday life in Cambridge and also prepare our target style images. In this application scenario, we select two breakthrough and well-known techniques - model pruning [2] and perceptual losses feed-forward transformation network [3] - to make our model smaller and faster. Our goal is to gain an understanding of how neural style transfer speed-up and model compression work in real practice.

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*King's College, University of Cambridge, CB2 1ST

2 Background

2.1 Style Transfer

The early 2000s saw the beginning of style transfer and texture transfer research. Hertzmann et al. [4] presented image analogies based on a straightforward multi-scale autoregression, whereas Efros and Freeman [5] proposed texture synthesis and transfer via image quilting. An iterative optimization technique for style transfer was put out by Gatys et al. [1] and makes use of the representations learned by convolutional neural networks. A pre-trained convolution neural network (CNN) is used in neural style transfer to transfer styles from one image to another. A loss function is constructed to reduce the discrepancies between generated images, style reference images, and content images. Since each step of the optimization problem necessitates a forward and backward trip over the pre-trained network, their method is computationally expensive. Johnson et al. [3] used a feed-forward network via perceptual loss and were able to speed up the image transform process in a single feed-forward pass. Although this network could produce results instantly, it is limited to the one style that it was trained on.

2.2 Model Pruning

Pruning a model often involves removing from the network any components that give little to no information during inference. Usually, a big initial network is trained to be accurate, and the aim is to create a smaller network with a similar level of accuracy [6].

To increase training speed, several algorithms prune the network during training [7, 8] or even at initialization [9]. Nevertheless, the majority of pruning algorithms adhere to the conventional method suggested by [2, 10] and perform pruning after the usual training procedure.

The literature review work [11] comes to the conclusion that intelligently allocating parameters to distinct layers [12, 2, 13, 14, 15] or global pruning [9, 16] likely to perform better than uniformly pruning all layers. Moreover, when keeping the number of fine-tuning iterations constant, many approaches create pruned models that perform better than initiating with the identical sparsity pattern [17, 18, 19, 16].

2.3 Evaluation of Neural Style Transfer

In the field of neural style transfer, there is always no single correct output. Currently, the most popular approach for evaluating neural style transfer models is a qualitative comparison of style and content images. A qualitative comparison is generally helpful for a first impression, however, the performance of style transfer models varies significantly across various styles and content images. A few works [20, 21] also employ classical perceptual metrics such as PSNR or SSIM [22], however, these metrics are generally not consistent with human perception [23]. Recently, Wright *et al.* [24] proposes a quantitative method by combining two factors: 1. content preservation between generated images X_g and content images X_c and 2.style feature distributions difference between generated images X_g and style images X_s. We adopt this metric for quantitative evaluation.

$$ArtFid(X_g; X_c; X_s) = (1 + \frac{1}{N} \sum_{i=1}^{N} d(X_c^{(i)}; X_g^{(i)})) \quad (1 + FID(X_s; X_g))$$

3 Methodology

As mentioned earlier, this project is to experience how model speed and size have been improved in this field. So we choose the basic and first algorithm in Neural Style Transfer [1] as the baseline. To witness the effect of model speed-up, we select the perceptual losses feed-forward transformation network [3] to implement. In addition, we choose one breakthrough paper [2] about model pruning and apply it to the speed-up model, to get a feel of size reduction. Ultimately, we apply this model in our proposed application scenario - real-life photo style transfer - and measure the changes in speed and size. It is worth mentioning that we did not include the most cutting-edge technology in this project. Because as beginners in this field, the first thing we want to do is to experience the basic principle and algorithms, to figure out the key development and improvement of this field.

Figure 1: Our Artist-Eye Net based on the framework of [3].

3.1 Baseline

We use the work of Gatys et al. [1] as our baseline. y_c and y_s are input content and style image. The network is a pre-trained VGG-16 [25] on ImageNet [26]. We use a pre-trained VGG-16 instead of VGG-19 for consistent with the network structure used in [3]. The style image y_s is passed through the network and its style representations on each layer are stored. The content representation of content image y_c is calculated on one layer and is also stored. The image y is initialized with white noise. y are updated iteratively until it can simultaneously match the style features of y_s and the content feature of y_c. An output image $\hat{y}$ evolves from y by minimizing the total loss in two aspects - content loss $\ell^{i}_{content}(y; y_c)$ and style loss $\ell^{j}_{style}{}^{2J}(y; y_s)$. i and j represent different network layers. $_c$ and $_s$ are scalars which represent weights for different loss items.

$$\hat{y} = \arg\min_{y} \; _c \ell^{i}_{content}(y; y_c) + \; _s \ell^{j}_{style}{}^{2J}(y; y_s)$$

Each iteration (the updation of y) requires a forward and backward pass through the VGG-16. A forward pass is needed to calculate the style and content loss, and then a backward pass is needed to calculate the total loss gradient - the loss derivative with respect to the pixels value - to update the image y's pixels. This makes the process slow. Thus, it is difficult to implement in production due to these time and computation constraints.

3.2 Artist-Eye Net

Our Artist-Eye Net is mainly based on the framework proposed by Johnson et al. [3]. This utilised a single feed-forward pass of the image transformation process. The limitation is that it is only capable of producing in the one style that it was trained on. But in our application scenario, this is acceptable. After taking pictures, users prefer to use the style templates that come with the photo editing software directly. Compared with the slow image generation speed, this limitation does not affect the user experience too much. We need to train multiple networks, each corresponding to a style template. Figure 1 shows how it works. The concept is to train an image transformation network to transform input images into output images in sigle feed-forward pass. The loss function of this image transferm net is also a deep convolutional network - a pretrained VGG-16 same as [1]. The differences between certain layers in the loss network can help to measure the loss in content and style. The output image $\hat{y}$ is measured by its content loss with the content image x (also act as input image), and its style loss with the style target image y_s.